

Homework solutions for 02YELMA

Chapter: Electric Fields in Matter

1. (a) The bound volume charge density is

$$\rho_b = -\nabla \cdot \mathbf{P}.$$

In cylindrical coordinates (radial symmetry),

$$\nabla \cdot \mathbf{P} = \frac{1}{s} \frac{d}{ds}(sP_s).$$

Since $P_s = \frac{a}{s}$,

$$sP_s = a \quad \Rightarrow \quad \frac{d}{ds}(a) = 0.$$

Therefore,

$$\boxed{\rho_b = 0 \quad \text{for } 0 < s \leq R.}$$

- (b) The bound surface charge density is

$$\sigma_b = \mathbf{P} \cdot \hat{n}.$$

At $s = R$, $\hat{n} = \hat{s}$, so

$$\boxed{\sigma_b = \frac{a}{R}.$$

- (c) Total bound charge per unit length:

$$Q_{\text{vol}} = \int \rho_b dV = 0.$$

Surface contribution (per unit length):

$$Q_{\text{surf}} = \sigma_b \cdot (2\pi R) = \frac{a}{R}(2\pi R) = 2\pi a.$$

Hence,

$$\boxed{\frac{Q_{\text{total}}}{\ell} = 2\pi a.}$$

- (d) According to what we calculated above, the total charge per unit length is not zero. This is not contradictory: although $\rho_b = 0$ everywhere in the bulk, charge can reside on the boundaries. The divergence theorem allows nonzero flux (surface charge) even if the divergence vanishes locally. The divergence being zero only applies in the region $0 < s < R$ but not at the boundaries. At one of the boundaries, $s = R$, $\mathbf{P}$ terminates abruptly, producing a surface charge. At the other boundary, $s = 0$, the divergence is not well-defined due to the $1/s$ singularity, which can be interpreted as a line charge along the axis. Thus, the net surface charge arises from the discontinuity in $\mathbf{P}$ at $s = R$ is compensated by the singular behavior at $s = 0$.

2. (a)

$$\rho_b = -\frac{dP_x}{dx} = -\frac{d}{dx}(kx^2) = -2kx.$$

$$\boxed{\rho_b(x) = -2kx.}$$

(b) Surface charge densities:

$$\sigma_b = \mathbf{P} \cdot \hat{n}.$$

At $x = +L$, $\hat{n} = +\hat{x}$:

$$\boxed{\sigma_b(+L) = kL^2.}$$

At $x = -L$, $\hat{n} = -\hat{x}$:

$$\boxed{\sigma_b(-L) = kL^2.}$$

(c) Volume contribution per unit area:

$$Q_{\text{vol}}/A = \int_{-L}^L (-2kx) dx = 0.$$

Surface contribution:

$$Q_{\text{surf}}/A = kL^2 + kL^2 = 2kL^2.$$

Hence,

$$\boxed{Q_{\text{total}}/A = 2kL^2.}$$

- (d) The system is *not* neutral; there is net positive bound charge. A polarization field with nonzero total bound charge cannot be created by internal dipole rearrangement alone in an initially neutral, isolated slab. To realize this configuration, you need either free charges supplied or removed at the boundaries (e.g. electrodes), or a process that effectively transports charge across the surface. The $\mathbf{P}$ still describes only dipoles locally (dipole moment per unit area). However, this particular form of $\mathbf{P}$ implicitly assumes that free charge has been supplied or removed to sustain it.
- (e) Since $\rho_b(x) = -2kx$, it is negative for $x > 0$ and positive for $x < 0$, increasing linearly in magnitude away from the center. Thus, charge density is larger in magnitude near the boundaries. Physically, this is because polarization grows with x^2 , so its spatial variation is strongest away from the center.

3. (a)

$$\rho_b = -\nabla \cdot \mathbf{P}.$$

In spherical coordinates:

$$\nabla \cdot (kr\hat{r}) = \frac{1}{r^2} \frac{d}{dr}(r^2 kr) = \frac{1}{r^2} \frac{d}{dr}(kr^3) = 3k.$$

Hence,

$$\boxed{\rho_b = -3k.}$$

Surface charge:

$$\boxed{\sigma_b = kR.}$$

(b) By symmetry, $\mathbf{E} = E(r)\hat{r}$.

Using Gauss's law inside:

$$Q_{\text{enc}} = \rho_b \frac{4}{3}\pi r^3 = (-3k)\frac{4}{3}\pi r^3 = -4\pi k r^3.$$

$$E(4\pi r^2) = \frac{Q_{\text{enc}}}{\varepsilon_0} \Rightarrow \boxed{E_{\text{in}} = -\frac{k}{\varepsilon_0}r.}$$

Outside:

$$Q_{\text{tot}} = (-3k)\frac{4}{3}\pi R^3 + (kR)(4\pi R^2) = 0,$$

so

$$\boxed{E_{\text{out}} = 0.}$$

(c) Potential at the center:

$$V(0) = \frac{1}{4\pi\varepsilon_0} \left(\int \frac{\rho_b}{r'} dV' + \int \frac{\sigma_b}{R} dA' \right).$$

(d) Volume contribution:

$$V_{\text{vol}} = \frac{1}{4\pi\varepsilon_0} (-3k) \int_0^R \frac{4\pi r'^2}{r'} dr' = \frac{-3k}{\varepsilon_0} \int_0^R r' dr' = \frac{-3k}{\varepsilon_0} \frac{R^2}{2}.$$

Surface contribution:

$$V_{\text{surf}} = \frac{1}{4\pi\varepsilon_0} \frac{kR}{R} (4\pi R^2) = \frac{k}{\varepsilon_0} R^2.$$

(e) Total:

$$V(0) = -\frac{3kR^2}{2\varepsilon_0} + \frac{kR^2}{\varepsilon_0} = \boxed{-\frac{kR^2}{2\varepsilon_0}}.$$

(f) The contributions do not cancel because the spatial weighting differs: volume charge is distributed throughout the sphere with $1/r'$ weighting, while surface charge is all at $r = R$. Even though total charge is zero, the potential need not vanish.